LAB 04

Introduction to **Object-Oriented Programming**

Inheritance

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Lab04 Exercises

❖ Lab04.1. Infinite Input of Student

Write a program to allow user to *infinitely input student* to a table by using *ArrayList*.
You can suggest Attributes (id, name, age, etc.)

Console:

```
Student #1:
Id: 111
Name: Tola
Age: 17

Do you want to add more (y/n)?: y
Student #2:
Id: 222
Name: Makara
Age: 16

Do you want to add more (y/n)?: y
Student #3:
Id: 333
Name: Kompheak
Age: 25

Do you want to add more (y/n)?: n
```

Console:

```
Do you want to add more (y/n)?: n

=================================================
| No      | ID      | Name              | Age     |
=================================================
| 1       | 111     | Tola              | 17      |
| 2       | 222     | Makara            | 16      |
| 3       | 333     | Kompheak          | 25      |
=================================================
```

Lab04 Exercises

❖ Lab04.2. Student management without using ArrayList

Write a program to allow user mange a list of student by using normal Array not ArrayList.
You can suggest Attributes (id, name, age, etc.)

Console:

```
==== Menu ===
1.  View all students
2.  Add a new student
3.  Delete a student
4.  Quit
Choose an option: 3

==== Delete a student ====
Input student ID: 11111
The following student has been deleted.
| 1       | 111     | Tola    | 17      |
```

Console:

```
==== Menu ===
1.  View all students
2.  Add a new student
3.  Delete a student
4.  Quit
Choose an option: 3

==== Delete a student ====
Input student ID: 2345
Student is not found. Try again
Input student ID: 3456
```

Lab04 Exercises

❖ Lab04.3. Book store management without using ArrayList

Write a program to allow user mange a list of book by using normal Array not ArrayList.
You can suggest Attributes (ISBN, title, price, etc.)

Console:

```
==== Menu ===
1.  View all books
2.  Add a new book
3.  Update a book
4.  Quit
Choose an option: 3

==== Update a book ====
Input ISBN: 11111
Please update the following:
ISBN: 22222
Title: Java codding first edition
Price: 0.99
Author: Unknown
```

Console:

```
==== Menu ===
1.  View all books
2.  Add a new book
3.  Update a book
4.  Quit
Choose an option: 3

==== Update a book ====
Input ISBN: 11111
Book is not found. Try again
Input ISBN: 11111
```

Lab04 Exercises

❖ Lab04.4. Video Search Engine without using ArrayList

Write a program to allow user mange and search a list of video by using normal Array not ArrayList.
You can suggest Attributes (Title, uploader, length, type, .etc)

Console:

```
==== Menu ===
1.  Search
2.  View all videos
3.  Add a new video
4.  Quit
Choose an option: 3

==== Video Search ====
Input title: Sin
3 videos found:
1. Prey Eh Kert by sin sisamuth
2. Pel Del Trov Yum - Sinn Sisamouth
3. Luoch Sneh Luoch Tuk - Sinn Sisamouth
```

Console:

```
==== Menu ===
1.  Search
2.  View all videos
3.  Add a new video
4.  Quit
Choose an option: 3

==== Video Search ====
Input title: Vanda
0 video found. Try again
Input title: G-Devid
```

Lab04 Exercises

❖ Lab04.5. Music Store

Write a program in Java to manage music store. Design your class follow the below class diagram:

Create a **Song** and **Album** and you can suggest attributes and behaviors:

Console: (Menu)

```
====== Menu ======
1.  View a music store
2.  Add a song
3.  Create an album
4.  Quit
Choose an option: 3
===== Create new album ====
Album title: Skull 2
Genre: Hiphop

Choose an option: 3
===== Create new album ====
Album title: Reborn
Genre: Hiphop
```

Console: (Option 2)

```
Choose an option: 2
===== Add a new song ====
Select following album:
1.  Skull 1
2.  Skull 2
3.  Reborn
Choose an opt: 1
Song title: Solo
Signer: Vannda
Length: 3mins
Price: 5

A new song added to the album
```

Console: (Option 1)

```
====== Music Store ======

Album: Skull 1
| 1 | Solo              | Vannda    | 3mins     | 5$        |
| 2 | QueenBee          | Vannda    | 5mins     | 12$       |

Album: Skull 2
None of song

Album: Reborn
| 1 | How about now     | G-Devith  | 2mins     | 10$       |
```

Lab04 Exercises

❖ Lab04.6. Storing a list of Student without using any Array

Write a program in Java to use only Class/Object concept to store infinite number of student without using Array or Arraylist (*Not allowed to use built-in array anywhere*). Follow the instruction below:

- ✓ Create a **Student** class with at least 5 attributes (id, name, gender, etc.)
- ✓ Create a **StudentList** class with following methods:
  - add (to add to last element of the list)
  - remove (to remove at a specific index)
  - removeLast (to remove last element from the list)
  - clear (to clear all elements from the list)

Usage example:

```java
public static void main(String[] args) {

    StudentList myArr = new StudentList();

    System.out.println("==> Add");
    myArr.add("Makara");
    myArr.add("Kompheak");
    myArr.display();
    System.out.println("\n==> Remove at last");
    myArr.removeLast();
    myArr.add("Minea");
    myArr.add("Mehsa");
    myArr.display();
    System.out.println("\n==> Remove at larger index");
    myArr.remove(10);
    System.out.println("==> Clear all element");
    myArr.clear();
    myArr.display();

}
```

Output

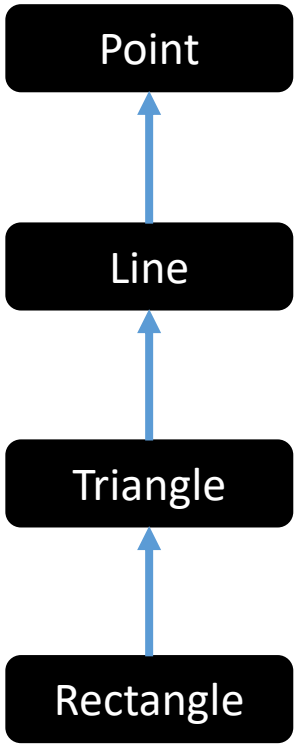
```
==> Add
Makara Kompheak
==> Remove at last
Makara Minea Mehsa
==> Remove at larger index
Index is out of range
==> Clear all element
Array is emply
```

Lab04 Exercises

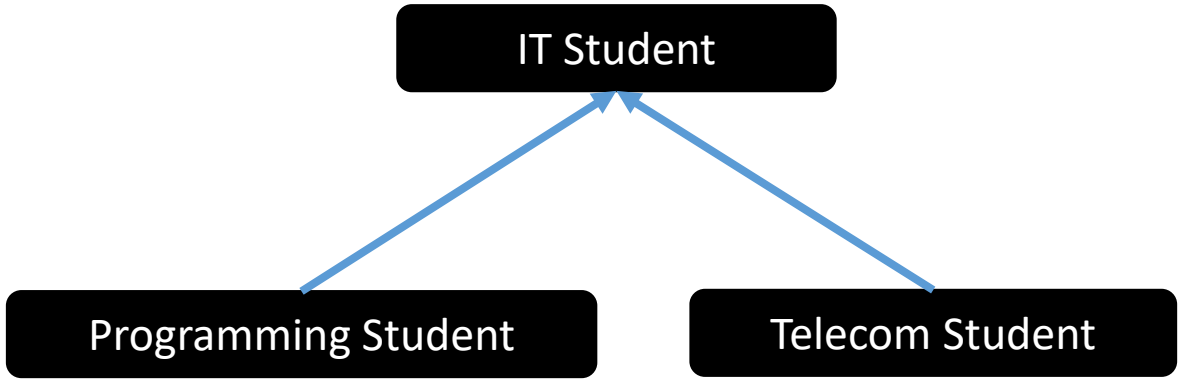
❖ Lab04.7. Inheritance in Java

Create a model following class diagrams and objects:

By following figure, implement classes following roles of multiple inheritance in Java with appropriate attribute/behavior to represent each class.

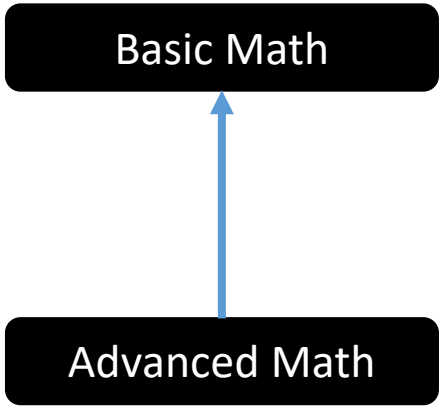


By following figure, implement classes following roles of hierarchical inheritance in Java with appropriate attribute/behavior to represent each class.



Lab04 Exercises

Follow the example in the figure to do your exercise 8:



Single inheritance

Implement classes in Java:

```java
class BasicMath {
    double addition(double a, double b) {
        return a + b;
    }
    double subtract(double a, double b) {
        return a + b;
    }
}

class AdvancedMath extends BasicMath {
    double multiply(double a, double b) {
        return a * b;
    }
    double divide(double a, double b) {
        return a / b;
    }
}
```

Usage:

```java
public class Demo4 {
    public static void main(String[] args) {
        BasicMath bm = new BasicMath();
        System.out.print(bm.addition(6, 9));
        System.out.print(bm.subtract(6, 9));

        AdvancedMath am = new AdvancedMath();
        System.out.print(am.addition(6, 9));
        System.out.print(am.subtract(6, 9));
        System.out.print(am.multiply(6, 9));
        System.out.print(am.divide(6, 9));
    }
}
```

Good luck 👌

✋ How to submit my Lab works???

👉 Screenshot your source codes and answers as a single **pdf** file and submit to the **Moodle** with Java file attachment as well.

Note: Don’t zip the files for submission